

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular Summer Examination –2023 Course: B. Tech. Branch : Electrical Semester :IV Subject Code & Name: BTEEPE405C Advance Renewable Energy Recourses Max Marks: 60 Date: 22-07-2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	State the types of energy sources. Also explain the advantages and disadvantages of renewable energy sources.	CO2	6
B)	Explain construction and working principle of proton exchange membrane fuel cell.	CO2	6
C)	What are the aspects of hydrogen as a fuel?	CO2	6
Q.2	Solve Any Two of the following.		
A)	With the help of neat block diagram explain wind energy conversion system.	CO2	6
B)	Explain types of vertical axis wind turbine with neat sketch.	CO2	6
C)	Explain the factors considered for appropriate site selection of wind farm.	CO2	6
Q. 3	Solve Any Two of the following.		
A)	Explain the electricity generation by photovoltaic effects in detail.	CO2	6
B)	Draw and explain I-V characteristics of solar PV cell. Also define fill factor.	CO2	6
C)	Explain equivalent circuit of solar PV cell with neat diagram. Also derive the equation, $V_{oc} = \frac{AKT}{q} \ln (I_L/I_S)$	CO2	6
Q.4	Solve Any Two of the following.		
A)	Explain different types of biomass conversion process.	CO2	6
B)	With the help of neat sketch, explain the classification of biogas plants.	CO2	6
C)	Explain in detail the small scale pyrolysis unit with the help of neat diagram.	CO2	6

Q. 5	Solve Any Two of the following.		
A)	Draw schematic diagram of lead acid cell and explain its working in detail.	CO1, CO2	6
B)	With the help of neat block diagram explain compressed air energy storage.	CO1, CO2	6
C)	Explain in detail thermal energy storage.	CO1, CO2	6
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